How Mediterranean Fruit Flies Resist Aging, Live Long and Remain Fertile

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Introduction

The Mediterranean fruit fly (*Ceratitis capitata*; medfly) is one of the most successful insects in the world with respect to its ability to invade new regions (Carey 1991). The species has spread from its origins in tropical Africa to virtually all areas of Mediterranean Europe, North Africa, and the Middle East, to most countries in South and Central America, to isolated pockets in North America (e.g., California and Florida), and to several regions in the Pacific, including the Hawaiian Islands and western Australia (Carey 1997).

One reason for the medfly's invasion success is that it is capable of persisting in small, sparsely distributed populations when hosts, food and mates are scarce. For example, entomologists frequently report the presence of extremely small medfly populations in regions where annual re-invasion is unlikely, including remote areas of Israel (Rivnay 1950), Hawaii (Vargas et al. 1983), California (Carey 1996), Australia (Maelzer 1990) and Greece (Papadopoulos et al. 1996). Inasmuch as these remote areas lack larval hosts for much of the year and medfly preadults (eggs-to-pupae) are not capable of diapause (Christenson and Foote 1960), one of the most important physiological adaptations that allow the medfly to persist at low population levels is the ability of the adult females to bridge unfavorable periods by reducing their rate of aging. Despite the importance of understanding how medflies survive in isolated regions with few resources, surprisingly few studies have been conducted on the physiological ecology of this species. And virtually no research has been published on the ecological context of aging in wild populations.

The underlying motive for this paper stems from the recent finding from laboratory studies in which medfly females were found to live longer and be capable of reproducing at later ages if they were maintained on a sugar-only diet at younger ages and then switched to a full diet at older ages (Carey et al., 1998). These findings are important because they suggest that there may be two physiological modes of aging – a waiting mode in which reproduction is low and survival high, and a reproductive mode in which reproduction and survival are both high initially but decrease rapidly at older ages. This concept more securely links the

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J.-M. Robine et al. (Eds.)
Research and Perspectives in Longevity
The Paradoxes of Longevity

aging literature (Finch 1991) with the physiological ecology literature concerned with the response of animals to periods of host, mate, and/or food shortages.

Baseline Medfly life Table Studies

The starting point for investigations on medfly aging was publication of the paper by Carey et al. (1992) reporting the results of the large-scale life table study of Mediterranean fruit flies (*Ceratitis capitata*). Age-specific mortality was monitored in 1.2 million medflies over their lifetimes at the Moscamed fruit fly rearing facility in Tapachula, Mexico. The main finding from this study was that death rates slowed at older ages, which implied that senescence cannot be characterized as an ever-increasing risk of death, that the Gompertz mortality model did not describe a unitary pattern or mortality at older ages in all species, and that there was no absolute life span limit.

A subtle feature of the mortality curve presented in this original paper, but one that was not addressed in it, was the appearance of a mortality "shoulder" (Fig. 1). The follow-up analysis of the data on the sex-specific mortality patterns (Carey et al. 1995 a), as well as a separate study on the effects of density on the

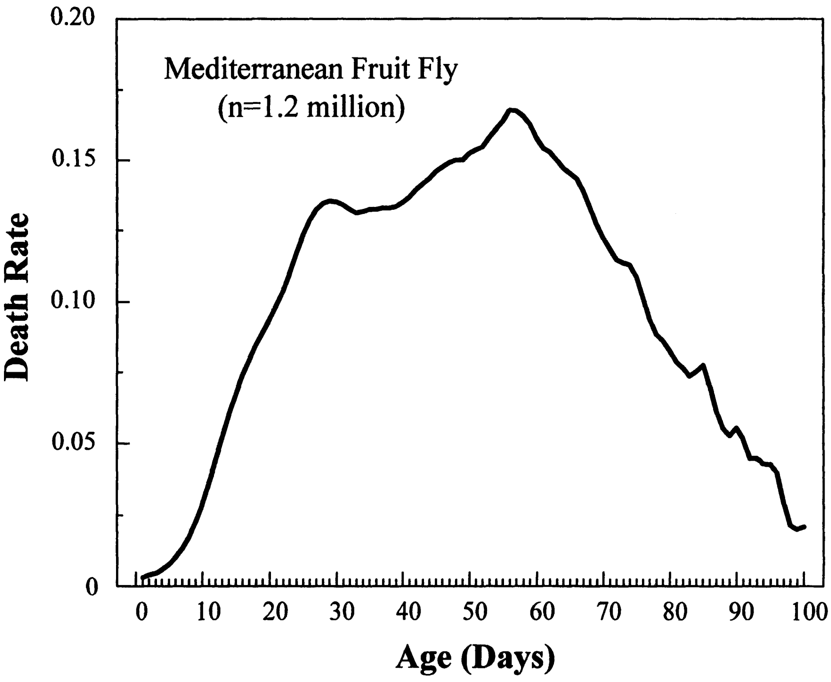

Fig. 1. Smoothed age-specific hazard rates medflies based on an initial number of 1.2 million individuals (both sexes). Note presence of the mortality "shoulder" at about 3 weeks (redrawn from Carey et al. 1992)

mortality patterns at older ages (Carey et al. 1995 b), revealed that this "shoulder" was expressed in female cohorts but not in male cohorts. The implication of this observation was that changes at the level of the individual (i.e., reproductive physiology) may account for changes in mortality trajectories at the level of the cohort.

This hypothesis, that individual-level changes in reproductive physiology explain the surge in mortality at younger ages in female cohorts, was tested by Müller et al. (1997). Experiments were conducted on medfly cohorts totaling over 400,000 flies maintained on one of two dietary regimes: 1) a sugar-plus protein diet in which flies were provided with sucrose and a protein source (referred to as a "full diet"); and 2) a protein-free diet in which flies were provided with a diet consisting only of sugar (protein-deprived). The main results from this study included: 1) life expectancy at eclosion was reduced in both male and female medflies if they were deprived of protein, though this reduction was greater in females (16.5 vs 12.1 days) than in males (15.2 vs 14.3 days); and 2) the sex differential in life expectancy favors females when cohorts are maintained on a full diet but reverses when flies are maintained on a sugar-only diet. As shown in Figure 2, the reversal of the male-female life expectancy differential is caused by a sustained surge in early female mortality under protein deprivation that is tied to egg-laying and physiological processes.

The finding that the surge in mortality at younger ages in sugar-fed female cohorts accounts for the significant difference in life expectancy between sugar-fed and protein-fed females, as well as for the reversal in sex-specific life expec-

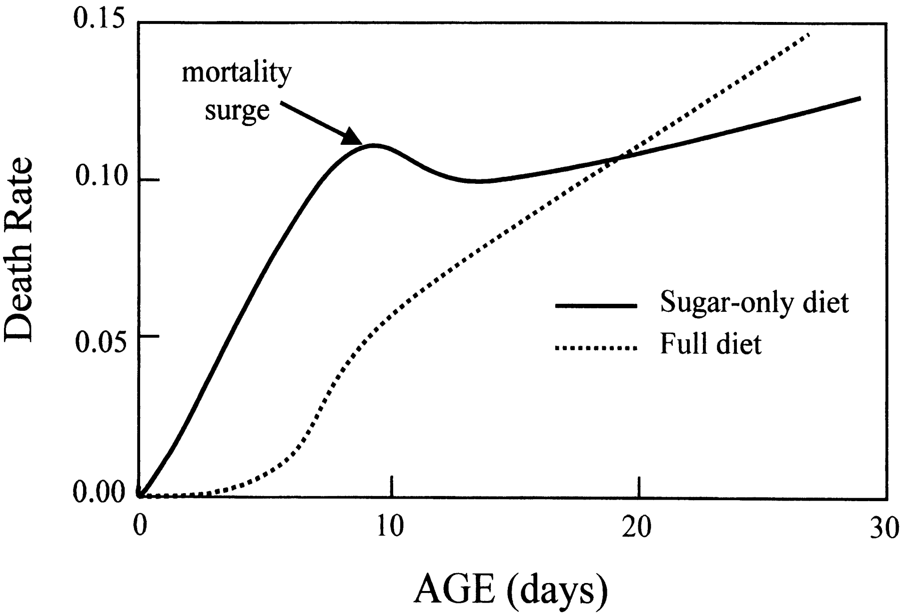

Fig. 2. Death rates for medfly females (n = 30,000 individuals) maintained on either sugar-only or a full diet (redrawn from Müller et al. 1997)

tancy, provides support for the initial hypothesis that individual-level changes in the reproductive physiology account for the mortality "shoulder" in female cohorts. This finding foreshadows the more recent discovery suggesting that medflies increase their survival rates by switching to a "waiting" mode of reproduction. In other words, the followup study by Müller et al. (1997) and a more recent one by Carey et al. (1998) suggest that the shoulder in the hazard curse of the original life table study of 1.2 million medflies (Carey et al. 1992), as well as the decline in mortality at the oldest ages, may be partly explained as due to the sugar-only diet on which the flies were maintained – the dietary regime caused the flies to remain in the "waiting" mode throughout their lives.

How Medflies Live Long and Remain Fertile

The ability of insects such as the medfly to enter arrested states and thus prolong survival and still retain the ability to reproduce evolved as an adaptation to synchronize the life cycle of animals with the rhythm of the environment and to insure that the active stages are present when conditions become favorable for reproduction (Andrewartha and Birch 1954; Mansingh 1971). Danks (1987) classified arrested states of animals as one of two types: 1) *predictive*, which refers to dormancy initiated in advance of the adverse conditions, such as hibernation in mammals or diapause in overwintering insects. In this case the conditions are variable but predictable; and 2) *consequential*, which is initiated in response to the adverse conditions themselves (e.g. germination of many plant seeds). This type of dormancy in animals is expected to evolve in environments that are unpredictable. In the following sections I describe and present data that reveal three ways in which medflies postpone their rate of aging as a *consequential* response to their environment: 1) by remaining a virgin – a consequence of not finding a mate; 2) by retaining matured eggs – a consequence of not having access to ovipositional hosts; and 3) by not manufacturing eggs – a consequence of not finding a source of dietary protein.

Virginity Prolongs Life

Many studies have documented the greater longevity of virgins over non-virgins, including studies on *Drosophila* (Smith 1958; Fowler and Partridge 1989; Partridge and Andrews 1985; Partridge 1986), the housefly, *Musca domestica* (Ragland and Sohal 1973), and the bruchiid beetle *Callosobruchus maculatus* (Tatar et al. 1993). To verify that this general principle also holds for medflies, we conducted large-scale life table studies on this species by gathering mortality data on 35,000 females maintained in all-female (virgin) cages and on 30,000 females maintained in mixed-sex (non-virgin) cages and fed a sugar-only diet (P. Liedo and J. R. Carey, unpublished data).

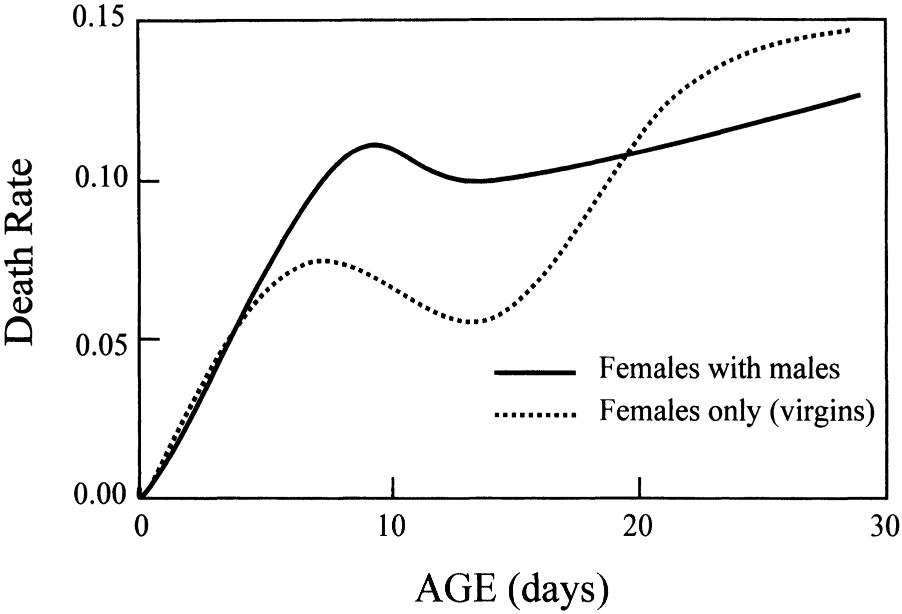

Fig. 3. Death rates of female medflies maintained in all-female (i.e., virgin) or mixed sex cages. Curves were computed using mortality data on 35,000 and 30,000 individuals for virgin and non-virgin cages, respectively (unpublished data from P. Liedo and J. Carey)

The result of these studies revealed that the life expectancy of females maintained in same-sex cages exceeded the life expectancy of females maintained in mixed-sex cages by 1.3 days (14.7 vs 16.1 days). This life expectancy differential was due to differences in mortality rates at younger ages. These rates reveal that death rates were approximately the same between virgin and non-virgin females for the first 5 days, at which time they diverged until the flies were nearly 3 weeks of age (Fig. 3). At this time the death rates for females in same-sex cages crossed over and exceeded the death rates for females maintained in mixed-sex cages. In general, the finding that the life expectancy at eclosion of unmated medfly females is greater than the life expectancy of mated females is consistent with the findings in virtually all other life table studies on virginity in insects (Reznick 1985; Bell and Koufopanou 1986).

Delayed or Intermittent Egg Laying Reduces Aging Rate

It is widely known that medflies often encounter periods when suitable hosts for oviposition are unavailable, ranging from a few days for an individual fly to find a scarce host to several months for entire populations to site and wait while hosts disappear and reappear due to seasonal changes (Drew et al. 1983; Fletcher et al. 1978; Meats and Khoo 1976). Because little is known about the demographic response of medflies to discontinuities in host availability, Carey et al. (1986)

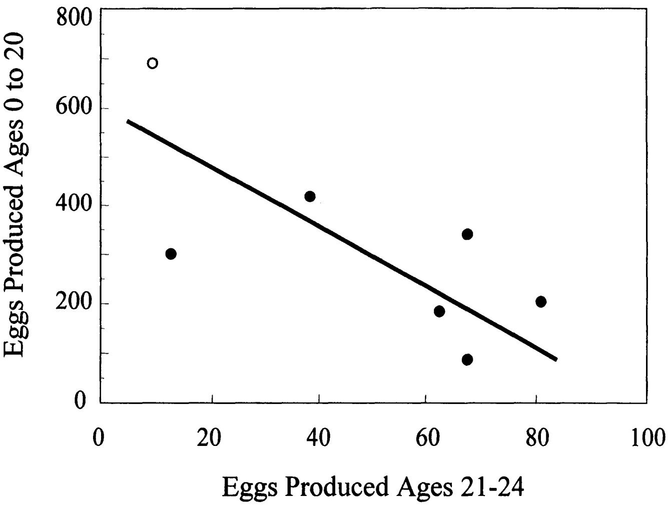

Fig. 4. Relationship between total reproduction during the first 20 days and reproduction from days 21 to 24 in medfly cohorts subjected to various host deprivation regimes. Open circle denotes control cohort in which females had access to hosts *ad libitum* (data from Table 2 in Carey et al. 1986)

conducted studies in which daily reproduction and survival were measured in cohorts of medflies subjected to periods of host deprivation. The design concept for this study was that two conditions were necessary to completely specify a period of host deprivation: 1) *level*, defined as the percentage of the total days in which the host is absent over a specified period; and 2) *pattern*, defined as the sequence of host days and host-free days over a specified period. The number of eggs produced by experimental females that had access to hosts early in the number produced by the 24 days of the study (control flies) was far fewer in the last 4 days than, females that were deprived of hosts at younger ages. For example, the control flies produced an average of 680 eggs/female during the first 20 days of the experiment but only 9.7 eggs/female in the last 4 days. By contrast, flies that were deprived of hosts for the first 16 days produced an average of 67 eggs/female during the last 4 days or around 6-fold more eggs than did the control flies. Overall the flies that were host-deprived at younger ages were capable of producing more eggs/day at older ages than were flies that had been laying throughout their lives, irrespective of the level and/or pattern of host deprivation. The number of eggs that were laid at early ages was inversely related to the number of eggs laid at older ages (Fig. 4) and directly related to total mortality after 24 days (Fig. 5). In general, the host deprivation study by Carey et al. (1986) revealed that: 1) medfly egg production is not only affected by a female's chronological age but also by her previous reproductive effort; and 2) although the cause of medfly egg production levels can be qualitatively separated into age effects and reproductive effects, no simply quantitative relationship exists between these two factors. In other words, it is not possible to assign a unit cost to future reproduction. Although a reproductive pacemaker may exist – individ-

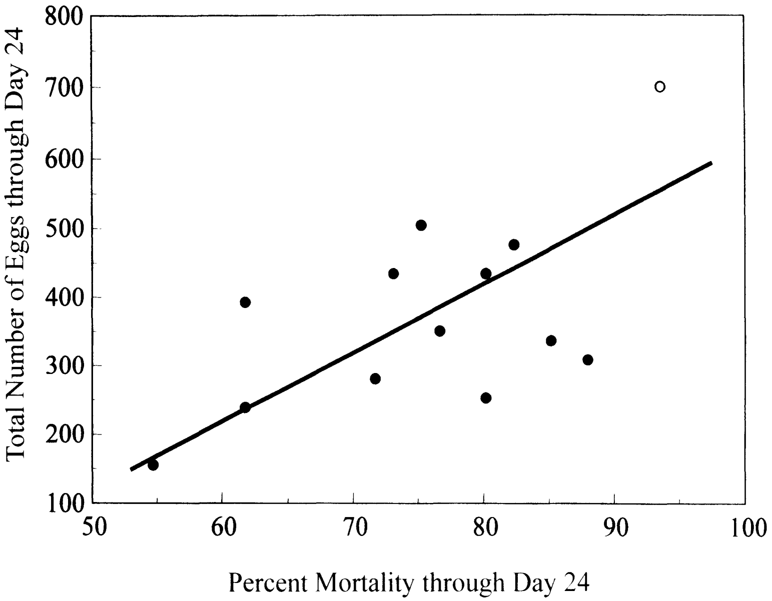

Fig. 5. Relationship between egg production in medfly cohorts subjected to various host deprivation regimes for 24 days and percentage mortality at the end of this period. Open circle denotes control cohorts in which females had access to hosts *ad libitum* (data from Tables 1 and 4 in Carey et al. 1986)

ual medfly females may only be endowed with a fixed number of germinal egg cells determined partly by genetics, partly by the environment and partly by chance – there are clearly other factors besides current and total reproductive output that determine longevity.

Protein Deprivation Suppresses Egg Production and Prolongs Life

Courtice and Drew (1984) studied the nutritional ecology of two fruit fly relatives of the medfly in Australia and concluded that "... food supply for egg production is the factor primarily responsible for fluctuations in numbers of *Dacus tryoni* and *D. neohumeralis*, the two major pest species in eastern Australia." Drew et al. (1983) stated "Our work suggests that nutrition, and particularly adult nutrition, is of overriding importance to the activity and numbers of fruit flies."

These findings provide context for the recent findings concerning dietary restriction of adult females by Carey et al. (1998), who concluded that the life history of medflies can be characterized by two physiological modes with different demographic schedules of fertility and survival: 1) a waiting mode, in which both mortality and reproduction are low; and 2) a reproductive mode, in which mortality is very low at the onset of egg laying but accelerates as eggs are laid. This discovery was the result of an experiment in which an initial pool of 2,500 adults of each sex was maintained in single pair cages with sugar and water and subgroups of 100 pairs were selected at 30, 60 and 90 days, provided with a full diet *ad libitum* and their reproduction and survival monitored until the last female died. Lifetime reproduction and survival were also monitored from eclosion in two 100-pair control cohorts – one on sugar-only and the other on a full diet. The following specific results point to this two-mode aging system: 1) the life expectancy of flies in all three treatments increased immediately on the days when they were given a full diet after having been maintained on a sugar-only diet; 2) the life expectancy of the full-diet control flies at eclosion was not only similar to the life expectancy of the treatment flies at the ages when they were first given a full diet, but also similar to their life expectancies 30 and 60 days after they were first given full diet, even though their absolute ages differed substantially; 3) the duration of the life-expectancy advantage in medflies maintained on a full diet was short relative to that of medflies maintained on a sugar-only diet. This general response of a short-term gain but a long-term reduction in life expectancy for flies switched from sugar to a full diet was evidenced in all three treatment cohorts; 4) female medflies maintained on a sugar-only diet and then switched to a full diet were capable of producing eggs at all ages tested, but lifetime egg production decreased as the age at the time of switch increased; 5) age-specific death rates for three of the 100-fly cohorts revealed striking similarities in their trajectories after they were given access to a full diet, even though their chronological ages differed by up to two months. They shifted to a different mortality "track" once their diet was switched from sugar to full and they began laying eggs.

The discovery that medflies experience two separate modes of aging is fundamental to biological, evolutionary and ecological research on aging because it provides a basis for replacing the unrealistic assumption of fixed demographic schedules in the Lotka equation, which is used as the basis for the evolutionary theory of aging, with the more realistic assumption that these schedules can be induced and thus are highly plastic. Research on the nature of the induced physiological shift that occurs when medflies are switched to protein may open a window into the fundamental processes that determine longevity (see Vaupel et al. 1998).

Discussion

Medflies are able to retain their youth at older ages when conditions are unfavorable for reproduction by remaining young physiologically even though they are old chronologically. Thus one of the demographic consequences of postponed aging is that chronological age becomes less relevant to the concepts of longevity and life span than does physiological age. Whereas physical (chronological) time is measured in hours, days, and years and is assumed to flow evenly and inexorably at the rate of solar time, physiological time does not pass at a constant rate through the time frame of physical time. Thus the more development that accumulates between two calendar dates, the faster time may be said to pass for the developing insect. Or, for sexually mature adults, the more eggs that are matured and deposited between two points in time, the more rapidly the insect may be said to age (Carrel 1931). An insect in the egg stage cannot become an adult without first passing through the larval and pupal stages although the stages through which it passes are, of course, independent of the development rate. Thus the direction and sequence of this development process is deterministic and it is most appropriately tracked on a physiological rather than on a chronological time scale.

The data on postponed aging in the medfly suggest that it may be useful to view insect aging and reproduction in the context of a physiological rather than a chronological time scale. Whereas the conventional Lexis diagram (Pressat 1985) consists of a series of life-lines representing individuals increasing in age in direct proportion to the increase in calender time, a modified life-line that follows a physiological time scale when an individual enters an arrested state will appear warped. The contrast between the two life-lines is shown in Figure 6 where the left-most linear curve depicts chronological age increasing at exactly the same rate as physical time but the right-most non-linear curve depicts physiological age changing at different rates relative to physical time. This schematic

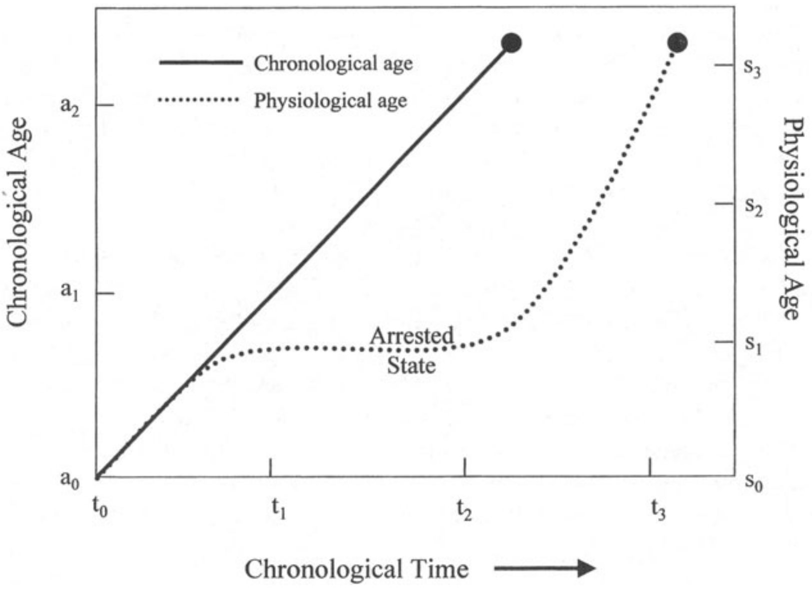

Fig. 6. Lexis diagram showing: 1) the conventional life line depicting the exact relationship between physical (calendar) time and chronological age – an organism's chronological age increases at the same rate as physical time. Chronological age, denoted a_i, passes at the same rate as physical time, denoted t_i; and 2) a physiological age life line where biological age is a non-linear function of chronological time. Physiological age/stage, denoted s_i, accumulates at different rates relative to physical time. An organism may enter an arrested state when conditions are unfavorable for growth or reproduction and thus its developmental and/or reproductive rate warps relative to its chronological age scale

illustrates why two individuals experiencing different environments – one where females enter an arrested stage and the other environment where females actively reproduce – will age at drastically different rates relative to chronological time.

The concept of physiological aging interpreted in a biogerontological context has several important implications with respect to aging and longevity. First, it renders invalid the concept of a fixed maximal age. This is because, unlike the chronological age of an individual, which progresses according to calendar time, the physiological age of an individual progresses according to the interaction between its physiological predisposition and its environment. Thus different conditions of host, food and mate availability will yield different physiological aging rates and thus different chronological life spans and maximal ages.

Second, up to now the evolutionary biology of aging was based on the unrealistic assumption that demographic schedules are fixed. The discovery that medflies experience two separate modes of aging is fundamental to biological, evolutionary and ecological research on aging because it provides a basis for replacing the former unrealistic assumption with the more realistic assumption that these schedules can be induced and thus are highly plastic. Thus research on the nature of the induced physiological shift that occurs when medflies are switched to protein may open a window into the fundamental processes that determine longevity. That demographic schedules can be induced will provide the empirical groundwork to stimulate development of a whole new class of demographic concepts, particularly those that assume that longevity is mediated through reproduction, which, in turn, is linked to the environment. Models that capture the interplay of birth and death rather than assume that these vital rates are independent will have a profound impact on the development of new theories of aging, as well as serve as touchstones for the emerging area of biodemography (see Carey and Gruenfelder 1997; Wachter and Finch 1997).

Third, by viewing the waiting mode of reproduction in the context of dormancy, postponed aging can be situated on one end of a dormancy continuum ranging from quiescence, reproductive diapause and shallow torpor to hibernation, aestivation and cryptobiosis on the other. Waiting strategies for prolonging survival while maintaining reproductive potential have been extensively documented in the physiological, ecological and natural history literature. The genes that regulate the transition to the waiting mode and survival in this mode are closely linked to longevity in nematode worms and yeast (Jazwinski 1996; Finch and Tanzi 1997). Thus research on the waiting mode and transitions in and out of it may provide key insights for understanding how many species remain young and retain their ability to reproduce at older ages in the wild.

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